19 Implementation of Naive Bayes classification for Bank Loan prediction

Code:

```
import pandas as pd
from sklearn.naive_bayes import GaussianNB
from sklearn.preprocessing import LabelEncoder

# Create dataset
data = {
    'Income': [30000, 40000, 50000, 60000, 80000, 90000],
    'Credit_Score': [550, 600, 650, 700, 750, 800],
    'Loan_Amount': [200000, 250000, 300000, 350000, 400000, 450000],
    'Loan_Status': ['No', 'No', 'No', 'Yes', 'Yes', 'Yes']
}

df = pd.DataFrame(data)

# Split input and output
X = df[['Income', 'Credit_Score', 'Loan_Amount']]
y = LabelEncoder().fit_transform(df['Loan_Status'])

# Train model
model = GaussianNB()
model.fit(X, y)

# Predict loan approval
new_applicant = [[70000, 720, 320000]]
prediction = model.predict(new_applicant)

# Display result
if prediction[0] == 1:
    print("Loan Approved")
else:
  print("Loan Rejected")
```

Output:

```
Gradient Boosting Classification
Training Accuracy: 100.0 %
Study Hours: 4
Attendance: 72 %
Predicted Result: Pass
```

